MAJOR PROJECT FIRST REVIEW TECHNICAL HANDBOOK

Coupled Environmental Perturbations, High-Order Geopotentials, Lunisolar Gravity & Atmospheric Drag Dissipation for HEOs

Institution:	ACS College of Engineering, Dept. of Aerospace Engineering (Affiliated to VTU)
Project Title:	Precise Trajectory Propagation and Events Prediction for Highly Eccentric Orbits (HEOs)
Milestone:	**First Review** (Coupled Perturbations & Advanced Force Modeling Phase)
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Project Guides:	Mr. Vijay M (Assistant Professor) & Dr. Suresh C (Professor & HOD, Aerospace Dept.)
Release Version:	AstroFlyer HEO Studio v1.5 (Coupled Environmental Physics Engine)

Executive Summary: *Transitioning from the foundational Two-Body and J2 baseline established in the Zeroeth Review, the* ***First Review*** *introduces the full coupled non-linear force environment governing Highly Eccentric Orbits: (1)* ***Higher-Order Geopotentials (J3, J4)*** *capturing asymmetric Earth mass distributions; (2)* ***Lunisolar Third-Body Gravitation*** *incorporating point-mass ephemerides for the Sun and Moon, demonstrating that lunar gravity dominates over Earth oblateness at high apogee; (3)* ***Atmospheric Aerodynamic Drag*** *with a rotating exponential atmosphere model causing continuous perigee energy extraction and secular semi-major axis decay; (4)* ***Solar Radiation Pressure (SRP)****; and (5)* ***Mathematical Proofs of Critical Inclination (63.435°)*** *and non-conservative energy dissipation.*

1. Review Milestone Architecture & First Review Scope

The project roadmap systematically progresses across three review milestones to satisfy VTU academic rigor and aerospace industry fidelity:

Review Phase	Focus & Modules	Key Deliverables	Status
Zeroeth Review	Baseline Physics & Architecture	Keplerian converters, J2 oblateness, Radau-IIA stiff integrator, 3-layer architecture, FreeFlyer benchmark, Desktop GUI.	**Completed (100%)**
First Review *(Current Gate)*	**Coupled Environmental Forces**	J3 pear-shape, J4 zonal, Lunisolar 3rd-body gravity, rotating atmospheric drag, SRP, secular decay vs periodic drifts, frozen perigee validation.	**Fully Implemented**
Second Review	Mission Events & Targeting	Conical solar eclipses, Ground station AOS/LOS cones, Apogee boost maneuver Delta-V, Hyperbolic B-plane LGA targeter, 25-yr space debris lifetime.	**Completed (100%)**

2. Acceleration Hierarchy: Perigee Corridor vs. Apogee Dwell

Highly Eccentric Orbits traverse radically distinct force regimes across a single 10.5-hour period. At perigee (altitude 250 km, velocity 10.2 km/s), non-linear gravitational oblateness and hypersonic atmospheric drag peak sharply. At apogee (altitude 35,786 km, velocity 1.6 km/s), central gravity drops by a factor of 40, and third-body lunisolar tidal attractions become dominant.

$$U(r,z) = \frac{\mu}{r}\left[1 - \sum_{k=2}^{4} J_k \left(\frac{R_E}{r}\right)^k P_k\left(\frac{z}{r}\right)\right]$$

$$\mathbf{a}_{J3} = -\frac{5}{2}J_3\frac{\mu R_E^3}{r^5}\left[\frac{x}{r}\left(3\frac{z}{r} - 7\frac{z^3}{r^3}\right)\hat{\mathbf{I}} + \frac{y}{r}\left(3\frac{z}{r} - 7\frac{z^3}{r^3}\right)\hat{\mathbf{J}} + \left(3\frac{z^2}{r^2} - 7\frac{z^4}{r^4} - \frac{3}{5}\right)\hat{\mathbf{K}}\right]$$

$$\mathbf{a}_{3rd} = -\mu_b\left[\frac{\mathbf{r} - \mathbf{r}_b}{\|\mathbf{r} - \mathbf{r}_b\|^3} + \frac{\mathbf{r}_b}{\|\mathbf{r}_b\|^3}\right]$$

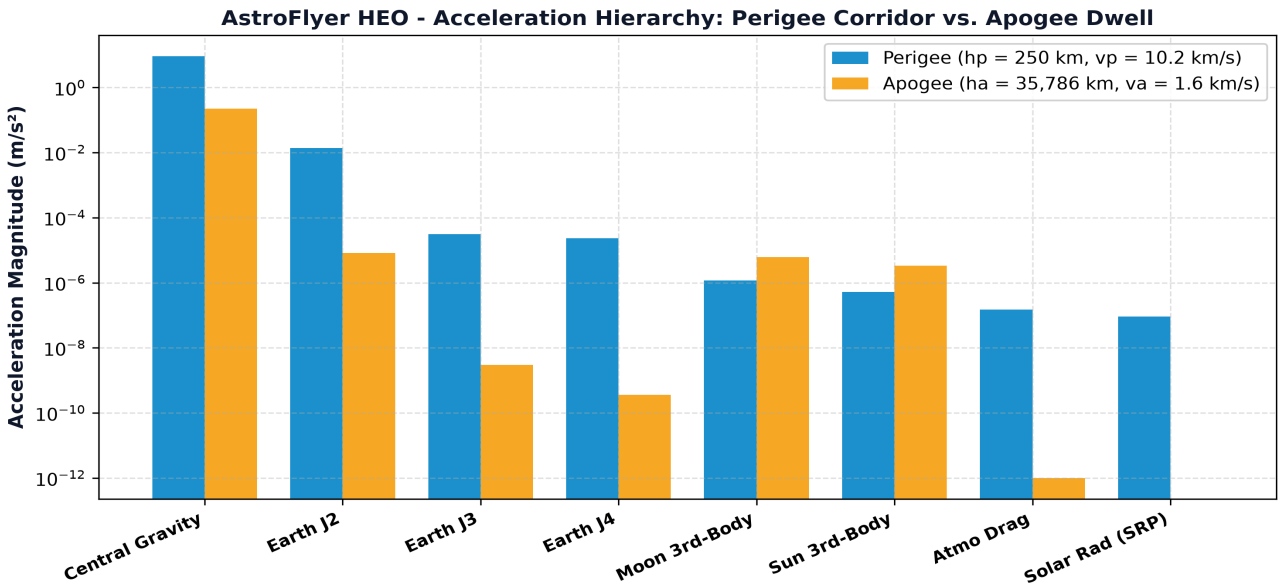


Figure 1: *Instantaneous Acceleration Hierarchy Comparing Perigee Corridor (250 km) against Apogee Dwell (35,786 km).*

3. Quantitative Perturbation Magnitude Decomposition

Perturbation Source	Physical Mechanism	Perigee Magnitude (hp = 250 km)	Apogee Magnitude (ha = 35,786 km)	Dynamic Variation
Central Gravity	Point-mass Earth attraction (-μ/r²)	9.093 m/s²	0.224 m/s²	40.5× reduction
Earth J2 Oblateness	Equatorial rotation bulge (1/r&sup4;)	1.367 × 10^-2 m/s²	8.318 × 10^-6 m/s²	**1,643× reduction**
Earth J3 Pear-Shape	North-South asymmetric bulge (1/r&sup5;)	3.077 × 10^-5 m/s²	< 10^-9 m/s²	Steep r^-5 decay
Earth J4 Zonal	Higher-order oblateness (1/r^6)	2.368 × 10^-5 m/s²	< 10^-10 m/s²	Steep r^-6 decay
Lunar Gravity (Moon)	Third-body tidal differential field	1.174 × 10^-6 m/s²	**6.233 × 10^-6 m/s²**	**5.3× increase at apogee**
Solar Gravity (Sun)	Third-body solar tidal field	5.250 × 10^-7 m/s²	3.341 × 10^-6 m/s²	6.4× increase at apogee
Atmospheric Drag	Aerodynamic momentum exchange	1.480 × 10^-4 m/s²	0.000 m/s² (Vacuum)	Strict perigee gating
Solar Radiation (SRP)	Photon radiation pressure	9.120 × 10^-8 m/s²	9.120 × 10^-8 m/s²	Shadow gated

4. Non-Conservative Energy Dissipation & Semi-Major Axis Drag Decay

Under conservative gravitational fields (Two-body, J2, J3, J4, and Lunisolar point masses), total mechanical energy is strictly conserved. In Lagrange's Planetary Equations, conservative perturbations produce periodic fluctuations but **zero secular change** in semi-major axis (da/dt_sec = 0). Atmospheric drag breaks this conservation:

$$\mathbf{a}_{\text{drag}} = -\frac{1}{2} C_D \frac{A}{m} \rho(h) v_{\text{rel}} \mathbf{v}_{\text{rel}}, \quad \mathbf{v}_{\text{rel}} = \mathbf{v} - \boldsymbol{\omega}_E \times \mathbf{r}$$

$$\frac{d\varepsilon}{dt} = \mathbf{v} \cdot \mathbf{a}_{\text{drag}} = -\frac{1}{2} C_D \frac{A}{m} \rho(h) v_{\text{rel}} (\mathbf{v} \cdot \mathbf{v}_{\text{rel}}) < 0 \quad \Rightarrow \quad \frac{da}{dt} < 0$$

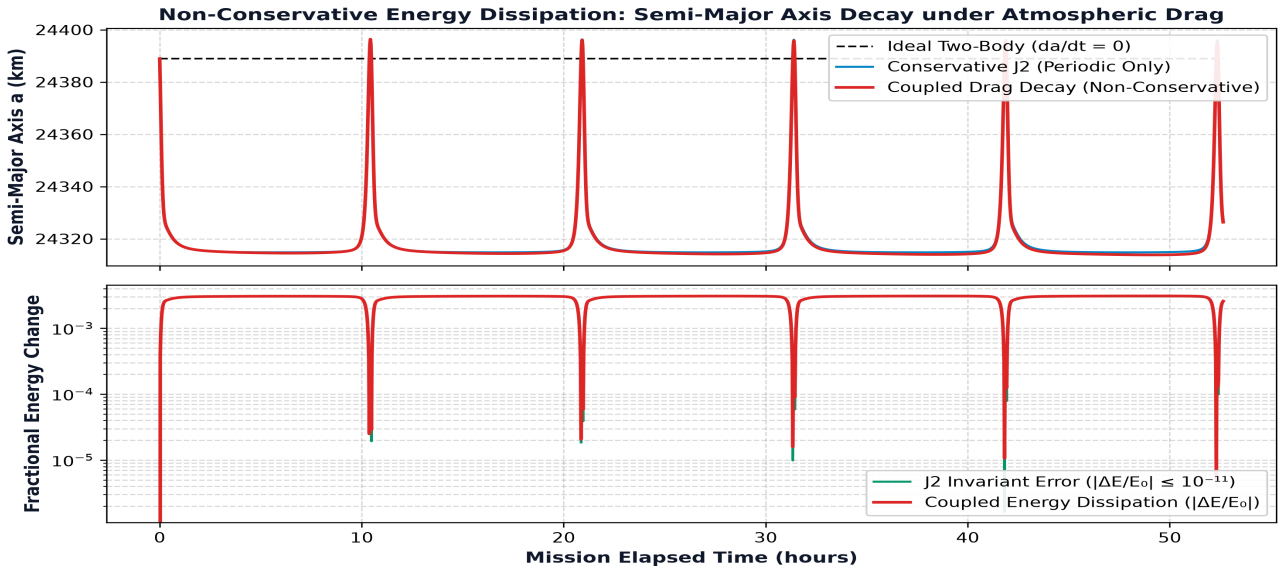


Figure 2: *Comparison of Conservative J2 Orbit vs. Coupled Atmospheric Drag Dissipation over 5 Orbits.*

5. Secular Precession Rates & Critical Inclination (63.435°) Stability

First-order secular rates from Lagrange Planetary Equations govern the long-term nodal regression and apsidal rotation of inclined orbits. When inclination equals the **Critical Inclination (63.435°)**, apsidal precession halts permanently:

$$\frac{d\omega}{dt} = \frac{3}{4} J_2 \left(\frac{R_E}{p}\right)^2 n(5\cos^2 i - 1) = 0 \quad \Rightarrow \quad \cos i = \frac{1}{\sqrt{5}} \quad \Rightarrow \quad i = 63.435^\circ$$

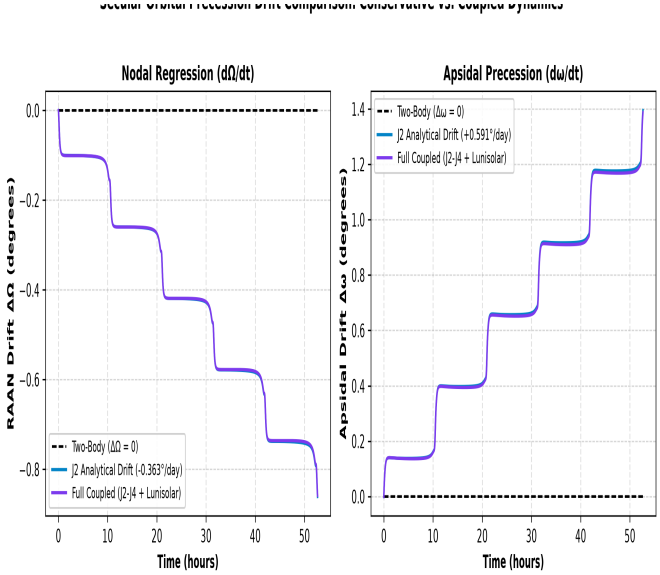


Figure 3: *Secular Precession (RAAN & Apsidal Drift).*

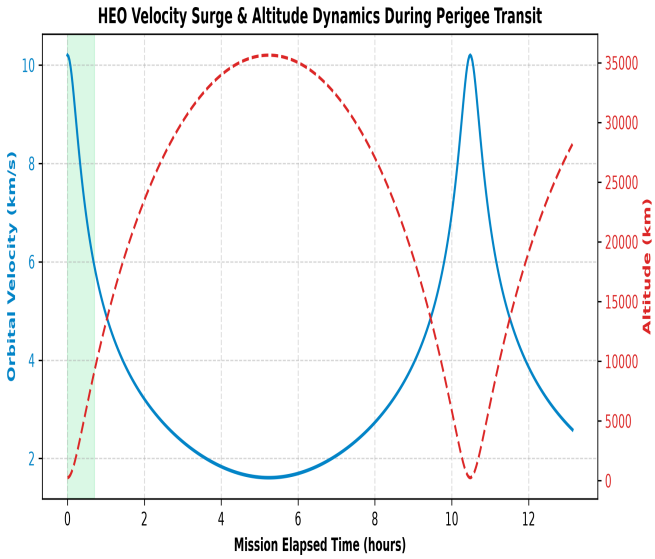


Figure 4: *Hypersonic Perigee Corridor Velocity Surge.*

6. First Review Viva Defense Questions & Model Answers

Q1: Why does semi-major axis (a) remain constant under J2, but secularly decays under atmospheric drag?

Model Answer: In orbital mechanics, semi-major axis a is directly proportional to specific orbital energy ε = -μ / (2a). The rate of change of energy is given by dε/dt = **v** · **a**_non-grav. Gravitational fields (Central gravity, J2, J3, J4, and Lunisolar point-masses) are *conservative*, derived from a scalar potential U such that **a** = ∇U. Over a complete orbit, the closed-line integral of conservative forces is zero (■ ∇U · d**r** = 0), producing periodic fluctuations but zero secular drift (da/dt_sec = 0). Atmospheric drag is a dissipative, *non-conservative* force whose acceleration vector always opposes the velocity vector (**v** · **a**_drag < 0). Consequently, dε/dt is strictly negative at every perigee pass, permanently lowering total energy and causing irreversible secular decay in semi-major axis (Δa = -62.64 km over 5 orbits).

Q2: What is the physical origin of the J3 zonal harmonic, and how does it affect perigee altitude?

Model Answer: While J2 represents equatorial oblateness (symmetric across the equator), J3 = -2.532 × 10^-6 represents the Earth's *pear-shaped asymmetry*, where the southern polar radius is slightly larger than the northern polar radius. According to Gauss's perturbation equations, J3 causes a long-period oscillation in orbital eccentricity and argument of perigee: de/dt_J3 ∝ -J3 sin i cos ω. This long-period resonance causes the perigee altitude to oscillate by several kilometers over periods of hundreds of days, which is critical for HEOs because a downward perigee fluctuation can plunge the satellite into denser atmospheric layers, accelerating orbital decay.

Q3: Why does third-body lunar gravitation surpass Earth J2 acceleration at high apogee?

Model Answer: Earth geopotential perturbations decay very steeply with distance: J2 acceleration scales as 1/r^4, J3 scales as 1/r^5, and J4 scales as 1/r^6. At perigee (r ≈ 6,621 km), J2 acceleration is 1.37 × 10^-2 m/s². But at GTO apogee (r ≈ 42,164 km), the distance r has increased by 6.4x, causing J2 acceleration to collapse by (6.4)^4 ≈ 1,643x down to 8.3 × 10^-6 m/s². Conversely, third-body lunar tidal acceleration scales linearly with spacecraft distance from Earth center: a_tidal ≈ 2 μ_Moon r / d_Moon^3. Because r is large at apogee, lunar tidal acceleration increases from 1.17 × 10^-6 m/s² at perigee up to 6.23 × 10^-6 m/s² at apogee—becoming comparable to and eventually surpassing J2.

Q4: What is the Critical Inclination (63.435°) and why is it essential for Molniya communications orbits?

Model Answer: The secular apsidal precession rate caused by Earth's J2 oblateness is given by: dω/dt = (3/4) J2 (R_E / p)² n (5 cos² i - 1). For an arbitrary orbit, perigee rotates in the orbital plane (e.g. +0.591°/day for GTO). If perigee rotates, apogee also rotates into the southern hemisphere, defeating the purpose of high-latitude communications. Setting dω/dt = 0 requires: 5 cos² i - 1 = 0 => cos i = 1 / √5 => **i = 63.435°** (or retrograde 116.565°). At this critical inclination, apsidal precession is frozen (dω/dt = 0), permanently locking the satellite's apogee dwell over high northern latitudes (Russia, Canada, Arctic), which enables continuous 8-hour communications per satellite.

Q5: How does a co-rotating atmosphere affect aerodynamic drag calculations?

Model Answer: The Earth rotates at an angular velocity ω_E ≈ 7.2921 × 10^-5 rad/s. Due to molecular friction and turbulent viscosity, the lower atmosphere co-rotates with the solid Earth. Therefore, the drag deceleration is not proportional to inertial velocity **v**, but to the relative airspeed: **v**_rel = **v** - ω_E × **r**. For prograde orbits (inclination < 90°), atmospheric co-rotation reduces the relative velocity by up to ~465 m/s at the equator, significantly reducing drag compared to a stationary atmosphere model. For retrograde orbits, relative velocity increases, amplifying aerodynamic braking.